J. Phase Equilib. Diffus. (2019) 40:653–667
https://doi.org/10.1007/s11669-019-00751-x

Thermodynamic Re-assessment of the Lanthanum-Tin System

Xiaojing Li[1] · Shuhong Liu[1] · Yuling Liu[1] · Yong Du[1]

Submitted: 31 January 2019 / in revised form: 11 July 2019 / Published online: 16 August 2019
© ASM International 2019

Abstract The lanthanum-tin system has been critically re-assessed based on available experimental data by the CALPHAD (CALculation of PHAse Diagram) method. The liquid phase was characterized by both the associated solution and substitutional solution models, while the seven intermetallics were treated as stoichiometric compounds. La_1Sn_1 was chosen as the associated complex in the associated solution model. The Redlich–Kister polynomial was used to describe the excess Gibbs energy of the substitutional solution model. Two sets of self-consistent thermodynamic parameters for the La-Sn system were finally obtained. The calculated results based on the thermodynamic parameters agree well with the phase diagram data and the thermodynamic data. Calculations using both sets of thermodynamic parameters obtained in the present work reproduce the experimental data reasonably well. Compared with previous assessments utilizing the associated solution and substitutional solution models, significant improvements have been made in the present work.

Keywords associated solution model · CALPHAD · La-Sn · substitutional solution model

✉ Shuhong Liu
shhliu@csu.edu.cn

[1] National Key Laboratory of Science and Technology for National Defence on High-strength Structural Materials, Central South University, Changsha 410083, Hunan, China

1 Introduction

Many AB_5-type hexagonal intermetallic compounds (where A represents a rare-earth metal and B is nickel or cobalt) have been reported to have considerable hydrogen absorption and desorption at room temperature.[1] Lanthanum (La) is one of the frequently used elements in rare-earth hydrogen storage materials. It has been stated that one formula unit of $LaNi_5$ can store approximately six hydrogen atoms.[2] Therefore, $LaNi_5$-based intermetallic compounds have been extensively investigated and it has been found that the addition of tin (Sn) can significantly improve the performance of $LaNi_5$ hydrogen storage properties.[1-7] Knowledge of phase equilibria and thermodynamic description is crucial for the application of a given material and for designing new materials. The CALPHAD (CALculation of PHAse Diagrams) method has shown efficient guidance on the research and design of materials.[8-10] Hence, to develop hydrogen storage materials, it is necessary to obtain an accurate thermodynamic database of the La-Ni-Sn-H system. To construct the entire quarternary La-Ni-Sn-H system, it is essential to have thermodynamic descriptions of all six constituent binary systems and four ternary systems. When we looked at the La-Sn system, we found that it needed a rigorous re-assessment, although it has been assessed by several groups.[11-13]

An extensive literature review of the La-Sn system was carried out by Palenzona and Cirafici.[14] Using the substitutional solution model for the liquid phase, Huang et al.[11] first optimized this system. As shown in Fig. 1(a), the calculated phase diagram from Huang et al.[11] showed an inverted miscibility gap, and the liquidus from 0 to 25 at.% La has not reproduced the experimental data well. Moreover, in their work, as an abrupt change of slope

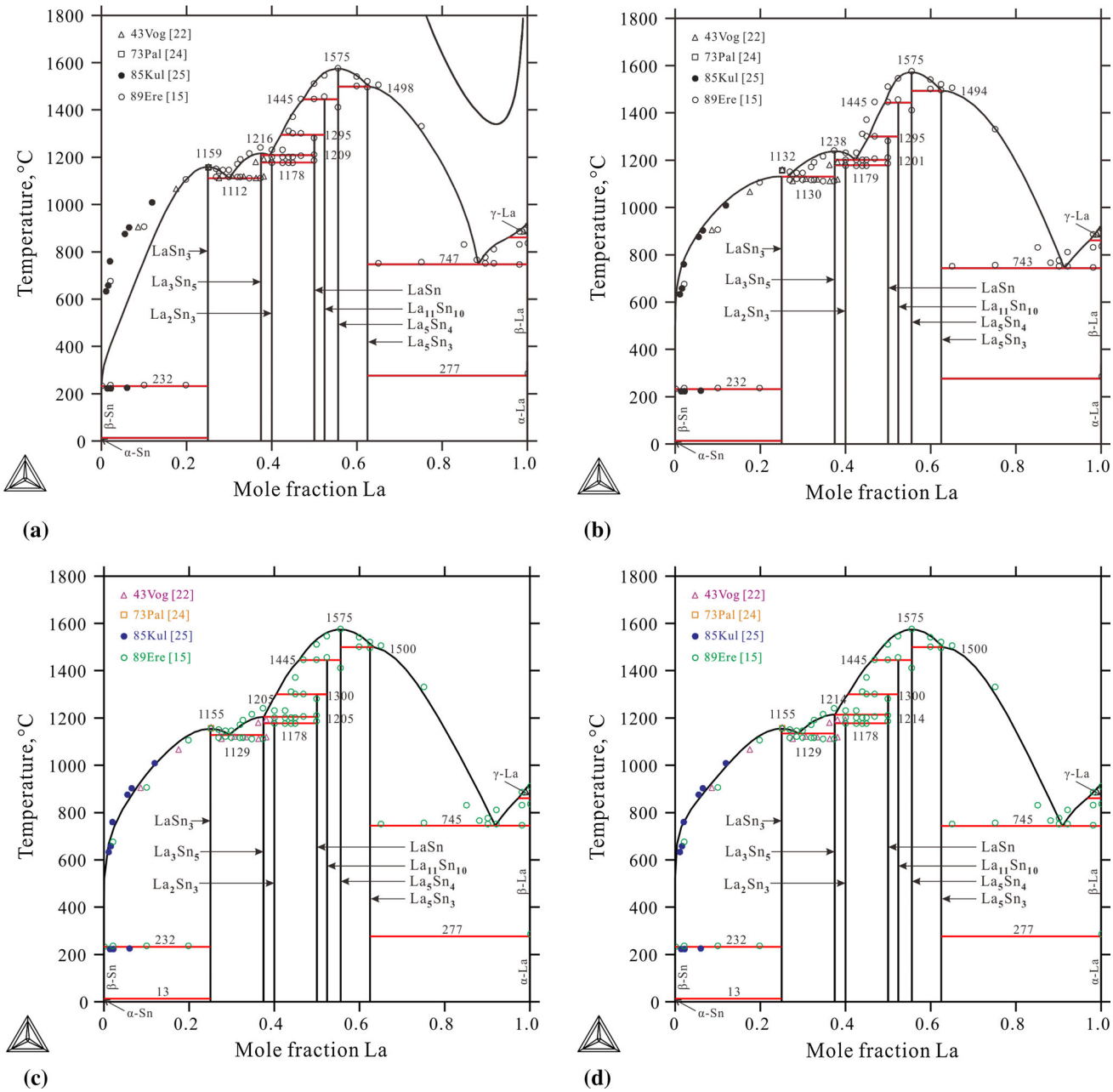

Fig. 1 (a) Calculated La-Sn phase diagram with the experimental data[15,22,24,25] by Huang et al.[11]; (b) Idbenali et al.[12]; (c) present work using an associated solution model (Set 1); and (d) present work using a substitutional solution model (Set 2)

existed in the heat capacity of pure Sn and the Neumann–Kopp relationship was used, the calculated heat capacity of $LaSn_3$ exhibited an abrupt change of slope around 232 °C, as shown in Fig. 3(a). Subsequently, Idbenali et al.[12] re-assessed the La-Sn system using the associated solution model for the liquid phase, and the liquidus slope in the range of 40–60 at.% La was too steep, as shown in Fig. 1 (b). The calculated enthalpies of mixing, the heat capacity of $LaSn_3$ and partial enthalpies of mixing using their thermodynamic parameters still presented a deviation from the experimental data, as shown in Fig. 2, 3(a) and 4(a). Recently, Kim et al.[13] re-assessed the La-Sn system using the modified quasichemical model for the liquid phase. The congruent melting of $LaSn_3$ and the reaction type of L+ LaSn=La_3Sn_5 calculated in their work were different from the experimental data.[15] Moreover, published thermodynamic parameters by Kim et al.[13] were incompatible with the thermodynamic databases developed by Thermo-Calc[16] or Pandat.[17]

Short-range order behavior in the La-Sn system has been reported by several groups.[18-20] With solution calorimetry (SC), some researchers found that the rare-earth metals in

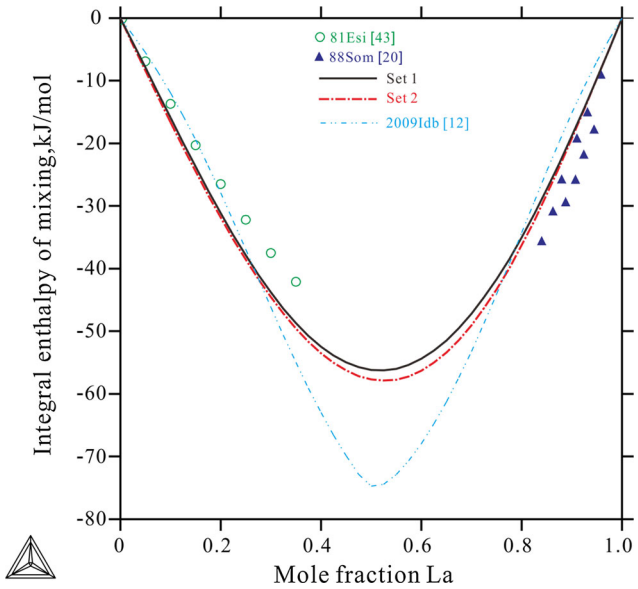

Fig. 2 Calculated enthalpies of mixing of the La-Sn system for liquid at 1127 °C using the obtained two sets of thermodynamic parameters with the experimental data[20,43] and previously calculated results.[11,12] The enthalpies of mixing data at different temperatures are converted into the same temperature, 1127 °C, via the function, $H^{\mathrm{exp}}_{\mathrm{mixing}}(T_{\mathrm{exp}}) + H^{\mathrm{cal}}_{\mathrm{mixing}}(1127\,^{\circ}\mathrm{C}) - H^{\mathrm{cal}}_{\mathrm{mixing}}(T_{\mathrm{exp}})$, where T_{exp} is the experimental temperature, $H^{\mathrm{exp}}_{\mathrm{mixing}}(T_{\mathrm{exp}})$ the measured enthalpies of mixing data at the experimental temperature, $H^{\mathrm{cal}}_{\mathrm{mixing}}$ (1127 °C) and $H^{\mathrm{cal}}_{\mathrm{mixing}}(T_{\mathrm{exp}})$ the calculated ones at 1127 °C and the experimental temperature, respectively

the liquid tin are highly exothermic, indicating that the bonding between rare-earth and tin atoms was quite strong.[18,19] This illustrates the existence of the associated complex in the La-Sn system. With experimental and calculated methods, Sommer et al.[20] confirmed the association was La_1Sn_1. Idbenali et al.[12] found that the calculated results would be more reasonable if La_1Sn_1 was used for the association in their re-assessment of the La-Sn system, although the composition of the associated complex in the liquid phase is usually based on the phase having the highest congruent melting temperature.

Considering the above facts, a thorough assessment of the La-Sn system is necessary. The present work is aimed to improve the CALPHAD-type thermodynamic description for the La-Sn system, using both the associated solution and substitutional solution models.

2 Literature Review

The available literature data of the La-Sn system, including phase diagram and thermodynamic properties data, have been critically reviewed in the present work and aresummarized in Tables 1 and 2.

2.1 Phase Diagram Data

Phase diagram data of the La-Sn system have already been published by several groups.[15,21-25] The phase diagram in

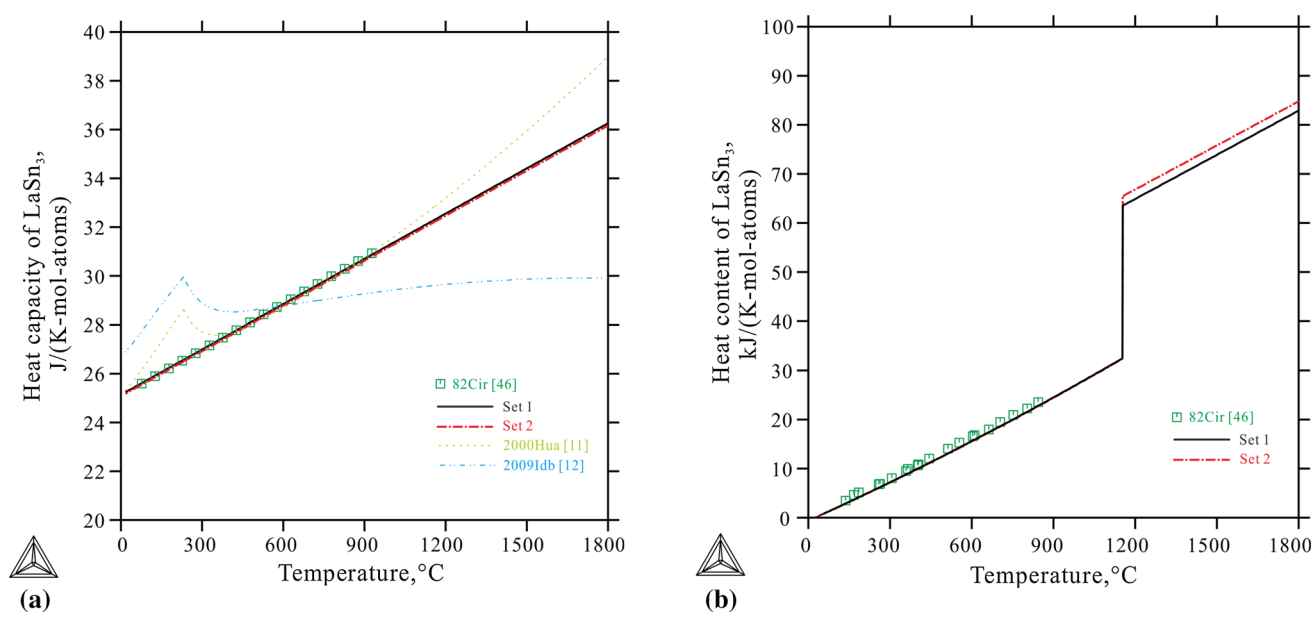

Fig. 3 (a) Calculated heat capacity of the $LaSn_3$ phase using the obtained two sets of thermodynamic parameters with the experimental data[46] and previously calculated results,[11,12] and (b) calculated heat content of the $LaSn_3$ phase using the obtained two sets of thermodynamic parameters with the experimental data[46]

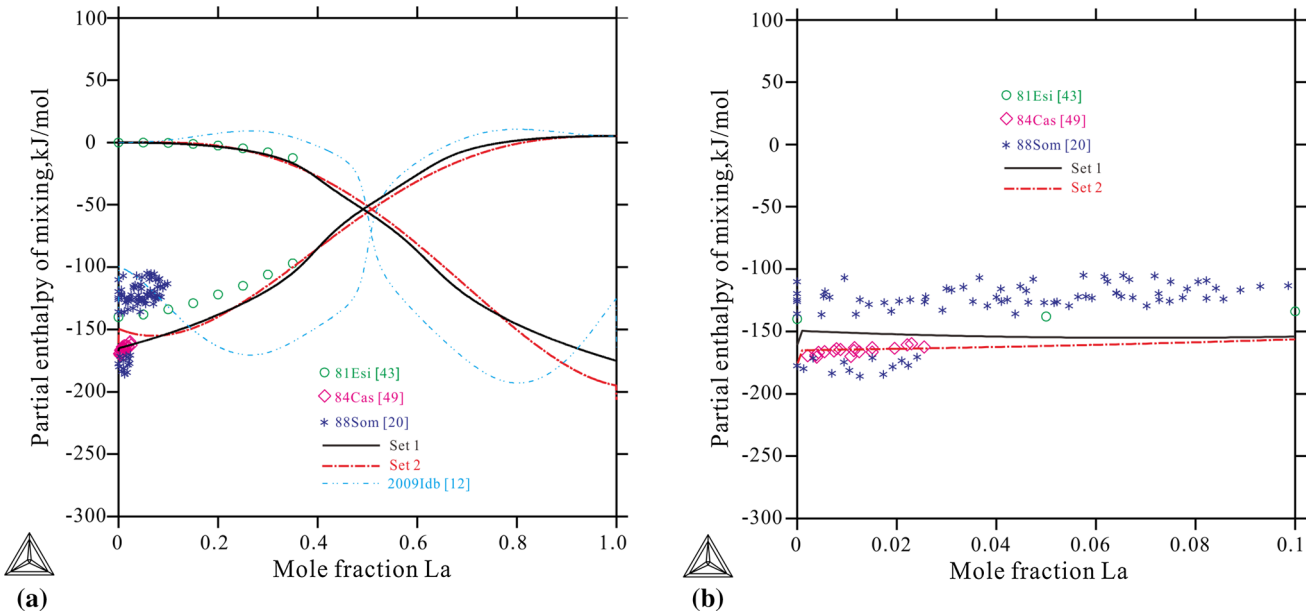

Fig. 4 Calculated partial enthalpies of mixing of the La-Sn system for liquid at 1127 °C using the obtained two sets of thermodynamic parameters with the experimental data[20,43,49] and previously calculated result[12]: (a) in the whole composition range of La and (b) in the range of 0–10 at.% La. The partial enthalpies of mixing data at different temperatures are converted into the same temperature, 1127 °C, via the function, $H^{exp}_{partial}(T_{exp}) + H^{cal}_{partial}(1127\,°C) - H^{cal}_{partial}(T_{exp})$ where T_{exp} is the experimental temperature, $H^{exp}_{partial}(T_{exp})$ the measured partial enthalpies of mixing data at the experimental temperature, $H^{cal}_{partial}$ 1127 °C and $H^{cal}_{partial}(T_{exp})$ the calculated ones at 1127 °C and the experimental temperature, respectively

the whole composition range was first measured by Canneri[21] using thermal analysis (TA). Due to the high impurity of the raw materials, the melting point of La was reported to be 812 °C by Canneri,[21] which is 108 °C lower than the data from SGTE.[26] Later, Vogel and Heumann[22] constructed the phase diagram in the composition range from 10 to 50 at.% La using TA. Kulagina et al.[25] investigated the liquidus around 0–25 at.% La by TA and microscopic observation (MO). Using MO, x-ray diffraction (XRD) and differential thermal analysis (DTA), Eremenko et al.[15] systematically re-determined the phase diagram data of the La-Sn system.[15] Considering the reliability and agreement of the data, phase diagram data by Kulagina et al.[25] and Eremenko et al.[15] were given a larger weight during the optimization.

The available crystal structure information of compounds are summarized in Table 1. The literature data on the phase diagram will be reviewed with intermetallics one by one.

The congruent melting point of $LaSn_3$ was investigated to be 1135, 1138, 1132±4, and 1155 °C by Vogel and Heumannl,[22] Palenzona,[24] Kulagina et al.[25] and Eremenko et al.[15] using TA, dynamic differential calorimetry (DDC) and DTA, respectively. As the data from Eremenko et al.[15] were systematic and their experimental method was reliable, their data were accepted in the present work.

Gambino et al.[23] noted that the La_3Sn_5 phase was a congruent melting compound. Later, Eremenko et al.[15] confirmed the melting point of La_3Sn_5 to be 1240 °C by DTA.

Using TA, the La_2Sn_3 phase was reported to be a congruent melting compound by two groups[21,22] and its melting point was determined to be 1187 and 1185 °C, respectively. However, based on the data from XRD and DTA, Eremenko et al.[15] reported that the La_2Sn_3 was formed by a peritectoid reaction: $LaSn+La_3Sn_5=La_2Sn_3$ at 1178±7 °C in the range from 37.5 to 50 at.% La using DTA. The early data of Canneri[21] and Vogel et al.[22] may not represent the actual phase equilibria because of the low purity of the starting metal and limited experimental techniques. Therefore, data from Eremenko et al.[15] were considered in the present optimization.

In addition, Eremenko et al.[15] also confirmed that the LaSn phase was formed by a peritectic reaction at 1295 °C using DTA.

The $La_{11}Sn_{10}$ phase was proposed by Eremenko et al.[15] according to its structure and composition and referring to the published $R_{11}Sn_{10}$.[27] It was reported to be formed at 1445 °C via $L+La_5Sn_4=La_{11}Sn_{10}$.[15]

Gambino et al.[23] reported that the La_5Sn_4 phase was congruent melting compound, and then Eremenko et al.[15] determined the melting point of La_5Sn_4 to be 1575 °C by DTA.

Table 1 Crystal structure data for the reported solid phases in the La-Sn system

Phase	Personal symbol	Prototype	Space group	Lattice parameters, nm			References
				a	*b*	*c*	
$LaSn_3$	cP4	$AuCu_3$	$Pm\bar{3}m$	0.4782			32
				0.4782			60
				0.4758	…	…	61
				0.47737	…	…	23
				0.4771	…	…	62
				0.4781	…	…	63
				0.478	…	…	35
				0.472	…	…	46
				0.4770	…	…	30
				0.478	…	…	25
				0.473	…	…	64
				0.4769	…	…	15
La_2Sn_5		Ce_2Sn_5	Cmmm	0.46233	3.57425	0.46817	36
				0.46385	3.5813	0.46983	37
La_3Sn_7		Ce_3Sn_7	Cmmm	0.45643	2.61558	0.46710	36
		Gd_3Sn_7		0.45783	2.6284	0.46799	37
$LaSn_2$		$NdSn_2$	Cmmm	0.44267	1.58552	0.45136	36
La_3Sn_5	oC32	Pu_3Pd_5	Cmcm	1.035	0.829	1.063	30
				1.035	0.829	1.061	15
La_2Sn_3	$P\bar{1}$			0.6518	0.8592	0.11296	47
LaSn	oC8	CrB	Cmcm	0.4782	1.194	0.4422	15
				0.47633	1.1911	0.44089	65
$La_{11}Sn_{10}$	tI84	$Ho_{11}Ge_{10}$	I4/mmm	1.20898	…	1.8166	65
La_5Sn_4	oP36	Sm_5Ge_4	Pnma	0.8448	1.626	0.8604	27
				0.845	1.624	0.861	30
				0.8429	1.619	0.8624	15
β–La_5Sn_3	hP16	Mn_5Si_3	$P6_3$/mcm	0.9416	…	0.6926	29
				0.9435	…	0.6961	28
				0.945	…	0.6975	30
α–La_5Sn_3	tI32	W_5Si_3	I4/mcm	1.2749	…	0.6343	31
				1.2748	…	0.6344	30
				1.2724	…	0.629	15
La_2Sn		…	…	0.650	…	0.636	66
La_3Sn	cP4	$AuCu_3$	Pm3 m	0.498	…	…	35

A La_5Sn_3 phase of the Mn_5Si_3 type was published by two groups,[28,29] and was named βLa_5Sn_3 in later work.[30] Gambino et al.[23] reported that La_5Sn_3 was formed congruently. Later, Franceschi[31] and Borzone et al.[30] reported that there are two allotropic structures of La_5Sn_3 (αLa_5Sn_3 and βLa_5Sn_3). Subsequently, Eremenko et al.[15] re-measured the structure of La_5Sn_3, but only αLa_5Sn_3 was observed, and the melting point of this phase was determined to be 1500 °C by DTA. Since no thermodynamic data on the α-β transformation for La_5Sn_3 have been published, we only consider the αLa_5Sn_3 phase reported by Eremenko et al.[15] in this work.

Besides the seven compounds mentioned above, there are five more intermetallic phases that have been reported by some researchers but were not certified to be stable by subsequent studies. One $LaSn_2$ phase was first reported by Canneri,[21] but it was confirmed to be $LaSn_3$ by Rossi.[32] There is no further report on the stability of $LaSn_2$. The La_3Sn phase has been investigated by several groups.[30,33-35] Based on these available data, we regarded the La_3Sn phase as a stable phase at high pressure. The La_2Sn_5 phase has been reported by several groups,[30,36,37] but neither phase diagram data nor thermodynamic data for this phase have been published. The crystal structure of

Table 2 Summary of the experimental data in the La-Sn system

Type of experimental data	Experimental technique	*T*, °C	Quoted mode (a)	References
Liquidus from 0 to 100 at.% La	TA	…	☒	20
Liquidus from 8.5 to 43.5 at.% La	TA	…	☒	22
Liquidus from 0 to 25 at.% La	TA	…	☑	25
Liquidus from 0 to 100 at.% La	MO, XPA, DTA	…	☑	15
Congruent melting point of $LaSn_3$	TA	1135	☒	22
Congruent melting point of $LaSn_3$	DDC	1138	☒	24
Congruent melting point of $LaSn_3$	DTA	1132±4	☒	25
Congruent melting point of La_3Sn_5	TA	1187	☒	21
Congruent melting point of La_3Sn_5	TA	1185	☒	22
Congruent melting point of $LaSn_3$, La_3Sn_5, La_5Sn_4	DTA	1155, 1240, 1575	☑	15
Eutectic temperature	TA	…	☒	20
	MO, XPA, DTA	1113(b), 1200(c), 747(d)	☑	15
Peritectic temperature	MO, XPA, DTA	235(e), 1445(f), 1295(g), 1500(h)	☑	15
Peritectoid temperature	MO, XPA, DTA	1179(i)	☑	15
Integral enthalpies of mixing	HC	1617	☑	43
	HC	901, 677	☑	20
Heat capacity of $LaSn_3$	…	−273 to 27	☒	44
	CHC	−268 to 27	☒	45
Heat capacity of La_2Sn_3	CDC	127–927	☑	46
	SAHP	−271.1 to −233	☒	47
Heat content of $LaSn_3$	CDC	127–927	☑	46
Partial enthalpies of mixing	SC	358–469	☒	19
	SC	452, 477, 502	☒	48
	HC	1617	☑	43
	SC	901	☑	49
	IC, CTC	677	☑	50
	DRC	657, 856	☑	51
	HC	825–1199	☑	20
Activities	EMF	400–700	☑	54
Enthalpies of formation for compounds	CA	0	☒	53
	CA	0	☒	48
	EMF	0	☑	54
	DDC	25	☑	24
	IC	27	☑	52
	EMF	727	☑	55
	IC	27	☑	30
	CA, CAL	…	☑	50
	DC	25	☑	51
Gibbs energies of formation	EMF	500	☑	55
	EMF	500, 727	☑	57

Table 2 continued

Type of experimental data	Experimental technique	*T*, °C	Quoted mode (a)	References
Entropies of formation	EMF	500	☑	55
	EMF	500, 727	☑	57

TA thermal analysis, *MO* microscopic observation, *XRD* x-ray diffraction, *DTA* differential thermal analysis, *DDC* dynamic differential calorimetry, *HC* high-temperature calorimeter, *SC* solution calorimeter, *IC* isoperibol calorimeter, *CTC* calvet-type twin calorimeter, *EMF* electromotive force, *CAL* calculation, *DC* dissolution calorimetry, *CHC* continuous heating calorimeter, *DC* dissolution calorimetry, *CDC* conventional drop calorimeter, *DRC* direct reaction calorimeter, *SAHP* semi-adiabatic heat-pulse

(a) Indicates whether the data are used or not in the parameter optimization: ☑ used; ☒ not used

(b) L=La_3Sn_5+$LaSn_3$

(c) L=LaSn+La_3Sn_5

(d) L=(βLa) +La_5Sn_3

(e) L+$LaSn_3$=(βSn)

(f) L+$La_{11}Sn_{10}$=LaSn

(g) L+La_5Sn_4=$La_{11}Sn_{10}$

(h) L+La_5Sn_4=La_5Sn_3

(i) LaSn+La_3Sn_5=La_2Sn_3

La_3Sn_7 and $LaSn_2$ was measured by Weitzer et al.[36] Dürr et al.[37] also determined the crystal structure of La_3Sn_7. However, none of these phases (La_2Sn_5, La_3Sn_7, $LaSn_2$) were confirmed by the ensuing experimental measurement of ternary systems including La and Sn.[38-42]

Therefore, only seven compounds, $LaSn_3$, La_3Sn_5, La_2Sn_3, LaSn, $La_{11}Sn_{10}$, La_5Sn_4 and αLa_5Sn_3, were considered in the present assessment.

2.2 Thermodynamic Data

Using a high-temperature calorimeter (HTC), the enthalpies of mixing of the liquid phase were investigated by Sommer et al.[20] and Esin et al.[43] at 855, 1010 and 1617 °C, respectively. These data were considered in the present optimization as they were obtained by a reliable method.

Using continuous heating calorimeter (CHC) and conventional drop calorimetry (CDC), the heat capacity of $LaSn_3$ was reported by Lethuillier and Chaussy,[44] Costa et al.[45] and Cirafici et al.[46] at −273 to 27, −268 to 27 and 127-927 °C, respectively. Using the semi-adiabatic heat-pulse method (SAHP), Fornasini et al.[47] obtained the heat capacity of La_2Sn_3 between −271.2 and −233 °C. Using CDC, Cirafici et al.[46] also investigated the heat content of $LaSn_3$ in the temperature range of 127-927 °C. Since we mainly focus on the data above 25 °C, the data from Cirafici et al.[46] were adopted in the present assessment.

The partial enthalpies of mixing at different temperatures were measured by several groups.[19,20,43,48-51] Using liquid–metal SC, Pratt and Morris[19] reported the partial enthalpies of mixing data at 359, 415 and 469 °C with the composition of La less than 0.1 at.%. Using the same method, Guadagno et al.[48] obtained the partial enthalpies of mixing data of La in liquid Sn at 452, 477 and 502 °C. Using HTC, Esin et al.[43] measured the partial enthalpies of mixing of La in liquid Sn at 1617 °C with the composition of La from 0 to 35 at.%. Using direct reaction calorimetry (DRC), Castanet[49] investigated the partial enthalpies of La in liquid Sn at 901 °C with the content of La up to 2.6 at.%. Through isoperibol calorimeter (IC) and Calvet-type twin calorimeter (CTC), Colinet et al.[50,51] reported the partial enthalpies of mixing of La in infinite dilution at 722 °C,[50] 657 °C[51] and 856 °C[51] at very small composition range of La. Using HTC, Sommer et al.[20] measured the partial enthalpies of mixing of La in liquid Sn with the content of La up to 10 at.% in the temperature range from 825 to 1199 °C. The data from these groups[20,43,49] were used in the present assessment since they were consistent with each other, and their experimental methods were clearly described.

The experimental activities of La in the liquid at 400, 500, 600 and 700 °C were reported by Matigorova and Bayanov[54] based on the measured data from the electromotive force (EMF). These data were evaluated by Kim et al.[13] and will be taken into consideration in the present optimization.

Many authors[24,30,48,50-55] have reported the enthalpies of the formation of compounds, and most of them are compatible. Using the calorimeter (CA) and SC, Guadagno et al.[48,53] determined the enthalpies of formation of $LaSn_3$, La_2Sn_3 and LaSn at 0 °C. Later, Matigorova and

Bayanov[54] re-investigated the enthalpies of the formation of $LaSn_3$ at 0 °C based on the measured data from EMF. The enthalpy of the formation of $LaSn_3$ at 25 °C was reported by Palenzona,[24] Colinet[51] and Kober et al.[55] using (DDC), DC and based on the data measured by EMF, respectively. Additionally, the enthalpy of the formation of La_5Sn_3 at 25 °C derived from EMF measurements was also reported by Kober et al.[55] Borsese et al.[52] measured the enthalpy of La_5Sn_3 at 27 °C by DC. Shortly after, Borzone et al.[30] reported the enthalpies of the formation of La_3Sn, La_5Sn_3, La_5Sn_4, La_3Sn_5, $LaSn_3$ at 27 °C using IC. In addition, Colinet et al.[50] calculated the enthalpies of the formation of compounds using Miedema's relationship.[56] Since data from Guadagno et al.[48,53] were too negative compared with other experimental data,[24,30,50-52,54,55] these data were not used in the present work. As the data from Borzone et al.[30] and Colinet et al.[50] were obtained for various compositions and were consistent with each other, these data were given a larger weight during the optimization.

Based on the measurement by EMF, the Gibbs energies and entropies of compounds were reported by two groups.[55,57] These data were considered in the present assessment as they were obtained by a relatively reasonable method and consistent with each other.

3 Thermodynamic Model

3.1 Pure Elements

The Gibbs energy functions of pure elements La and Sn were taken from the SGTE compilation by Dinsdale[26] and described by the following equation:

$$G_I(T) = H_i^{\mathrm{SER}} = A + BT + CT\ln T + DT^2 + ET^{-1} + IT^7 + JT^{-9} \quad \text{(Eq 1)}$$

where H_i^{SER} is the molar enthalpy of the element i at 1 bar and 25 °C in its standard element reference (SER) state, and T is the absolute temperature. The last two terms in Eq 1 are used only outside the ranges of the melting point, IT^7, for a liquid below the melting point and JT^{-9} for solid phases above the melting point.

3.2 Liquid Phase

The associated solution model was used to describe the La-Sn system as the short-range order behavior reported by several groups.[18-20] It was assumed that three species (La, La_1Sn_1, Sn) exist in the liquid phase. The Gibbs energy of the liquid phase for this model was described by Eq 2:

$$\begin{aligned} G_m^{\mathrm{liq}} - H^{\mathrm{SER}} = {} & y_{\mathrm{La}} \cdot {}^0G_{\mathrm{La}}^{\mathrm{liq}} + y_{\mathrm{Sn}} \cdot {}^0G_{\mathrm{Sn}}^{\mathrm{liq}} + y_{\mathrm{La_1Sn_1}} \cdot {}^0G_{\mathrm{La_1Sn_1}}^{\mathrm{liq}} \\ & + R \cdot T \cdot (y_{\mathrm{La}} \cdot \ln y_{\mathrm{La}} + y_{Sn} \cdot \ln y_{\mathrm{Sn}} \\ & + y_{\mathrm{La_1Sn_1}} \cdot \ln y_{\mathrm{La_1Sn_1}}) \\ & + y_{\mathrm{La}} \cdot y_{\mathrm{Sn}} \sum_{j=0}^{\infty} {}^jL_{\mathrm{La,Sn}}^{\mathrm{liq}} (y_{\mathrm{La}} - y_{\mathrm{Sn}})^j \\ & + y_{\mathrm{La}} \cdot y_{\mathrm{La_1Sn_1}} \sum_{j=0}^{\infty} {}^jL_{\mathrm{La,La_1Sn_1}}^{\mathrm{liq}} (y_{La} - y_{\mathrm{La_1Sn_1}})^j \\ & + y_{\mathrm{La_1Sn_1}} \cdot y_{Sn} \sum_{j=0}^{\infty} {}^jL_{\mathrm{La_1Sn_1,Sn}}^{\mathrm{liq}} (y_{\mathrm{La_1Sn_1}} - y_{\mathrm{Sn}})^j \end{aligned} \quad \text{(Eq 2)}$$

where ${}^0G_{\mathrm{La_1Sn_1}}^{\mathrm{liq}}$ is the binary interaction parameter, which is usually dependent on temperature:

$${}^0G_{\mathrm{La_1Sn_1}}^{\mathrm{liq}} = {}^0G_{\mathrm{La}}^{\mathrm{liq}} + {}^0G_{\mathrm{sn}}^{\mathrm{liq}} + a + bT \quad \text{(Eq 3)}$$

At the same time, the substitutional solution model is used to describe the liquid phase for comparison. The molar Gibbs energy of the substitutional solution model for the liquid phase is described by Eq 4:

$$\begin{aligned} G_m^{\mathrm{liq}} = {} & x_{\mathrm{La}} \cdot {}^0G_{\mathrm{La}}^{\mathrm{liq}} + x_{\mathrm{Sn}} \cdot {}^0G_{\mathrm{Sn}}^{\mathrm{liq}} + R \cdot T \\ & \cdot (x_{\mathrm{La}} \cdot \ln x_{\mathrm{La}} + x_{\mathrm{Sn}} \cdot \ln x_{\mathrm{Sn}}) \\ & + x_{\mathrm{La}} \cdot x_{\mathrm{Sn}} \sum_{i=0}^{\infty} {}^iL^{\mathrm{liq}} (x_{\mathrm{La}} - x_{\mathrm{Sn}})^i \end{aligned} \quad \text{(Eq 4)}$$

where ${}^iL^{\mathrm{liq}}$ is the binary interaction parameter of the La-Sn system, which is usually dependent on temperature:

$${}^iL^{\mathrm{liq}} = a + bT \quad \text{(Eq 5)}$$

3.3 Stoichiometric Compounds

The intermediate phases $LaSn_3$, La_3Sn_5, La_2Sn_3, LaSn, $La_{11}Sn_{10}$, La_5Sn_4, and La_5Sn_3 in the La-Sn system were treated as stoichiometric compounds since no solid solubility has been reported. Since there is measured heat capacity data of $LaSn_3$, it is preferable to express its Gibbs energy relative to the SER state. Therefore, the Gibbs energy of $LaSn_3$ per mole-formula is expressed as:

$$G_m^{\mathrm{La_ASn_B}} - H_{\mathrm{La}}^{\mathrm{SER}} - 3 \cdot H_{\mathrm{Sn}}^{\mathrm{SER}} = a + bT + cT\ln T + dT^2 + eT^{-1} \quad \text{(Eq 6)}$$

When there is no measured heat capacity available for compounds, the Gibbs energy per mole of formula unit La_ASn_B (except for $LaSn_3$) can be presented using the Neumann–Kopp rule as:

Table 3 Presently obtained thermodynamic parameters for the La-Sn system

Phase	Thermodynamic model	Parameters
Associated solution model (Set 1)		
Liquid	(La, La_1Sn_1, Sn)	${}^0G^{liq}_{La_1Sn_1} - {}^0G^{liq}_{La} - {}^0G^{liq}_{Sn} = -116873 - 9.89T$
		${}^0L^{liq}_{La,Sn} = -98695 - 2.02T$
		${}^0L^{liq}_{La,La_1Sn_1} = -58185 - 4.17T$
		${}^0L^{liq}_{La_1Sn_1,Sn} = -53284 - 20.14T$
$LaSn_3$	$La_{0.25}Sn_{0.75}$	${}^0G^{LaSn_3}_{La:Sn} = -68105.81 + 103.92T - 23.44T\ln T - 0.0031T^2$
La_3Sn_5	$La_{0.375}Sn_{0.625}$	${}^0G^{La_3Sn_5}_{La:Sn} - 0.375H^{SER}_{La} - 0.625H^{SER}_{Sn} = -83942 + 4.82T$
La_2Sn_3	$La_{0.4}Sn_{0.6}$	${}^0G^{La_2Sn_3}_{La:Sn} - 0.4H^{SER}_{La} - 0.6H^{SER}_{Sn} = -86008.6 + 5.18T$
LaSn	$La_{0.5}Sn_{0.5}$	${}^0G^{LaSn}_{La:Sn} - 0.5H^{SER}_{La} - 0.5H^{SER}_{Sn} = -89000 + 2.99T$
$La_{11}Sn_{10}$	$La_{0.524}Sn_{0.476}$	${}^0G^{La_{11}Sn_{10}}_{La:Sn} - 0.524H^{SER}_{La} - 0.476H^{SER}_{Sn} = -88635 + 2.15T$
La_5Sn_4	$La_{0.556}Sn_{0.444}$	${}^0G^{La_5Sn_4}_{La:Sn} - 0.556H^{SER}_{La} - 0.444H^{SER}_{Sn} = -87487 + 1.47T$
La_5Sn_3	$La_{0.625}Sn_{0.375}$	${}^0G^{La_5Sn_3}_{La:Sn} - 0.625H^{SER}_{La} - 0.375H^{SER}_{Sn} = -77448 - 0.72T$
Substitutional solution model (Set 2)		
Liquid	(La, Sn)	${}^0L^{liq}_{La,Sn} = -224650 - 20.01T$
		${}^1L^{liq}_{La,Sn} = -20200 + 22.06T$
		${}^2L^{liq}_{La,Sn} = +49802.6$
$LaSn_3$	$La_{0.25}Sn_{0.75}$	${}^0G^{LaSn_3}_{La:Sn} = -69832.4 + 104.13T - 23.39T\ln T - 0.0031T^2$
La_3Sn_5	$La_{0.375}Sn_{0.625}$	${}^0G^{La_3Sn_5}_{La:Sn} - 0.375H^{SER}_{La} - 0.625H^{SER}_{Sn} = -83366.3 + 3.8T$
La_2Sn_3	$La_{0.4}Sn_{0.6}$	${}^0G^{La_2Sn_3}_{La:Sn} - 0.4H^{SER}_{La} - 0.6H^{SER}_{Sn} = -84726.87 + 3.61T$
LaSn	$La_{0.5}Sn_{0.5}$	${}^0G^{LaSn}_{La:Sn} - 0.5H^{SER}_{La} - 0.5H^{SER}_{Sn} = -89085 + 2.11T$
$La_{11}Sn_{10}$	$La_{0.524}Sn_{0.476}$	${}^0G^{La_{11}Sn_{10}}_{La:Sn} - 0.524H^{SER}_{La} - 0.476H^{SER}_{Sn} = -88627 + 1.12T$
La_5Sn_4	$La_{0.556}Sn_{0.444}$	${}^0G^{La_5Sn_4}_{La:Sn} - 0.556H^{SER}_{La} - 0.444H^{SER}_{Sn} = -87487 + 0.36T$
La_5Sn_3	$La_{0.625}Sn_{0.375}$	${}^0G^{La_5Sn_3}_{La:Sn} - 0.625H^{SER}_{La} - 0.375H^{SER}_{Sn} = -81826.2 + 0.78T$

All parameters are given in J/mole and temperature (T) in °C. The Gibbs energies for the pure elements are from the SGTE compilation[26]

$$G_m^{La_ASn_B} - A \cdot H^{SER}_{La} - B \cdot H^{SER}_{Sn} = a + bT + A \cdot {}^0G^{dhcp}_{La} + B \cdot {}^0G^{bct_A5}_{Sn} \quad \text{(Eq 7)}$$

where a and b are the parameters to be evaluated in the optimization procedure. However, as mentioned recently by Schick et al.,[58] this expression may result in unphysical discontinuities of heat capacities, resembling second-order phase transitions. Concerning this problem, Schick et al.[58] proposed a modified Neumann–Kopp treatment of the heat capacity of stoichiometric phases by modified Cp at the melting point constant. However, the method by Schick et al.[58] was not used in this work since the basis of the problem is in the unary data for Sn not in the application of Neumann–Kopp itself.

3.4 Solid Solution Phases

The solid solution of (αSn), (βSn), (αLa), (βLa) and (γLa) phases in the La-Sn system were considered as unary phases since they have negligible solubility.

4 Optimization Procedure and Results

The present assessment was performed by Thermo-Calc software which works by minimizing the square sum of the errors.[59] To avoid the appearance of an inverted miscibility gap in the liquid phase, an additional constraint, QF (LIQUID)>1, was used in the liquid phase region during the optimization process. QF is a "state variable" in PARROT and POLY that has the value of the smallest eigenvalue of the determinant of all second derivatives of the Gibbs energy for all constituents. Based on the uncertainties of the data, a certain weight was given for each kind of experimental data. During the assessment, the parameters of the liquid phase were first optimized using the data of enthalpies of mixing,[20,43] partial enthalpies of mixing[20,43,49] and activity of La in liquid.[54] Later, the parameters in Eq 6 and 7 for stoichiometric compounds were optimized by considering the data of enthalpies of formation[24,30,50-52,54,55] and heat capacity[46] as well as the phase diagram.[15] Finally, all the parameters were

Table 4 Calculated invariant equilibria in the La-Sn system using two sets of parameters and compared with the literature data

Reaction	*T* (°C)	Type	Composition, La at.%			References
L=$LaSn_3$+(βSn)	232	Eutectic	0	0.25	0	11
	232		0	0.25	0	12
	232		0	0.25	0	13
	232		0	0.25	0	Set 1
	232		0	0.25	0	Set 2
L+$LaSn_3$=(βSn)	235±2	Peritectic	0.197	0.25	0	15
L=$LaSn_3$	1155	Congruent	0.25	0.25		15
	1159		0.25	0.25		11
	1131		0.25	0.25		12
	1138		0.25	0.25		13
	1155		0.25	0.25		Set 1
	1155		0.25	0.25		Set 2
L=La_3Sn_5+$LaSn_3$	1113±3	Eutectic	0.295	0.375	0.25	15
	1112		0.299	0.375	0.25	11
	1130		0.263	0.375	0.25	12
	1128		0.286	0.375	0.25	13
	1129		0.295	0.375	0.25	Set 1
	1135		0.290	0.375	0.25	Set 2
L=La_2Sn_3+$LaSn_3$	1108	Eutectic	0.315	0.4	0.25	67
L+LaSn=La_3Sn_5	1200	Peritectic	0.359	0.5	0.375	13
L=La_3Sn_5	1240	Congruent	0.375	0.375		15
	1216		0.375	0.375		11
	1238		0.375	0.375		12
	1240		0.375	0.375		13
	1205		0.375	0.375		Set 1
	1214		0.375	0.375		Set 2
L=LaSn+La_3Sn_5	1200±2	Eutectic	0.435	0.5	0.375	15
	1209		0.396	0.5	0.375	11
	1201		0.424	0.5	0.375	12
	1205		0.376	0.5	0.375	Set 1
	1214		0.375	0.5	0.375	Set 2
LaSn+La_3Sn_5=La_2Sn_3	1179±7	Peritectoid	0.5	0.375	0.4	15
	1178		0.5	0.375	0.4	11
	1179		0.5	0.375	0.4	12
	1176		0.5	0.375	0.4	13
	1178		0.5	0.375	0.4	Set 1
	1178		0.5	0.375	0.4	Set 2
L+$La_{11}Sn_{10}$=LaSn	1295	Peritectic	0.467	0.524	0.5	15
	1295		0.422	0.524	0.5	11
	1300		0.455	0.524	0.5	12
	1295		0.397	0.524	0.5	13
	1300		0.405	0.524	0.5	Set 1
	1300		0.406	0.524	0.5	Set 2

Table 4 continued

Reaction	T (°C)	Type	Composition, La at.%			References
$L+La_5Sn_4=La_{11}Sn_{10}$	1445	Peritectic	0.522	0.556	0.524	15
	1445		0.474	0.556	0.524	11
	1444		0.493	0.556	0.524	12
	1447		0.462	0.556	0.524	13
	1445		0.458	0.556	0.524	Set 1
	1445		0.466	0.556	0.524	Set 2
$L=La_5Sn_4$	1575	Congruent	0.556	0.556		15
	1575		0.556	0.556		11
	1575		0.556	0.556		12
	1578		0.556	0.556		13
	1575		0.556	0.556		Set 1
	1575		0.556	0.556		Set 2
$L+La_5Sn_4=La_5Sn_3$	1500	Peritectic	0.649	0.556	0.625	15
	1498		0.628	0.556	0.625	11
	1494		0.625	0.556	0.625	12
	1495		0.631	0.556	0.625	13
	1500		0.634	0.556	0.625	Set 1
	1500		0.629	0.556	0.625	Set 2
$L=(\beta La)+La_5Sn_3$	747±3	Eutectic	0.885	1	0.625	15
	747		0.885	1	0.625	11
	743		0.914	1	0.625	12
	751		0.910	1	0.625	13
	747		0.919	1	0.625	Set 1
	745		0.909	1	0.625	Set 2

Set 1 indicates the associated solution model; Set 2 indicates the substitutional solution model

optimized simultaneously based on the experimental data, and two sets of self-consistent thermodynamic parameters were obtained. These two sets of parameters for the La-Sn binary system are presented in Table 3. The calculated invariant equilibria in this system along with the experimental data and previous calculations are summarized in Table 4.

Figure 1(c) and (d) shows the calculated phase diagram using the two sets of parameters obtained in the present work with the experimental data,[15,22,24,25] respectively. It can be seen that the calculated phase diagram using both the associated solution model (Set 1, Fig. 1c) and substitutional solution model (Set 2, Fig. 1d) can reproduce with the experimental data reasonably. However, it is worth noting that, if we reproduce the congruent melting of La_3Sn_5 better, the liquidus slope would be much steeper near 40 at.% La. Although the liquidus proposed by Eremenko et al.[15] shows a steep slope near 40 at.% La, the slope is only established over a limited composition range. This is not likely to be described, since it requires a drastic change of Gibbs energy of liquid at a small composition range. Therefore, experimental data of the melting point of La_3Sn_5 may not reproduce the experimental data well in the present work.

Figure 2 illustrates the calculated enthalpies of mixing at 1127 °C, compared with the experimental data.[20,43] Enthalpies of mixing data at different temperatures has been converted into the same temperature, 1127 °C, via the function, $H^{\mathrm{exp}}_{\mathrm{mixing}}(T_{\mathrm{exp}}) + H^{\mathrm{cal}}_{\mathrm{mixing}}(1127\,^{\circ}\mathrm{C}) - H^{\mathrm{cal}}_{\mathrm{mixing}}(T_{\mathrm{exp}})$ where T_{exp} is the experimental temperature, $H^{\mathrm{exp}}_{\mathrm{mixing}}(T_{\mathrm{exp}})$ the measured enthalpies of mixing data at the experimental temperature, $H^{\mathrm{cal}}_{\mathrm{mixing}}(1127\,^{\circ}\mathrm{C})$ and $H^{\mathrm{cal}}_{\mathrm{mixing}}(T_{\mathrm{exp}})$ the calculated ones at 1127 °C and the experimental temperature, respectively. From Fig. 2, we found that the calculated results using Set 1 and Set 2 cannot reproduce the experimental data very well. During our optimization, we found that the data from Sommer et al.[20] is considered to be a little negative to reproduce the phase diagram. On the other hand, the data of Esin et al.[43] are considered to be a slight positive for reproducing the phase diagram. Therefore, considering the data from these two groups,[20,43] a more reasonable result has been obtained.

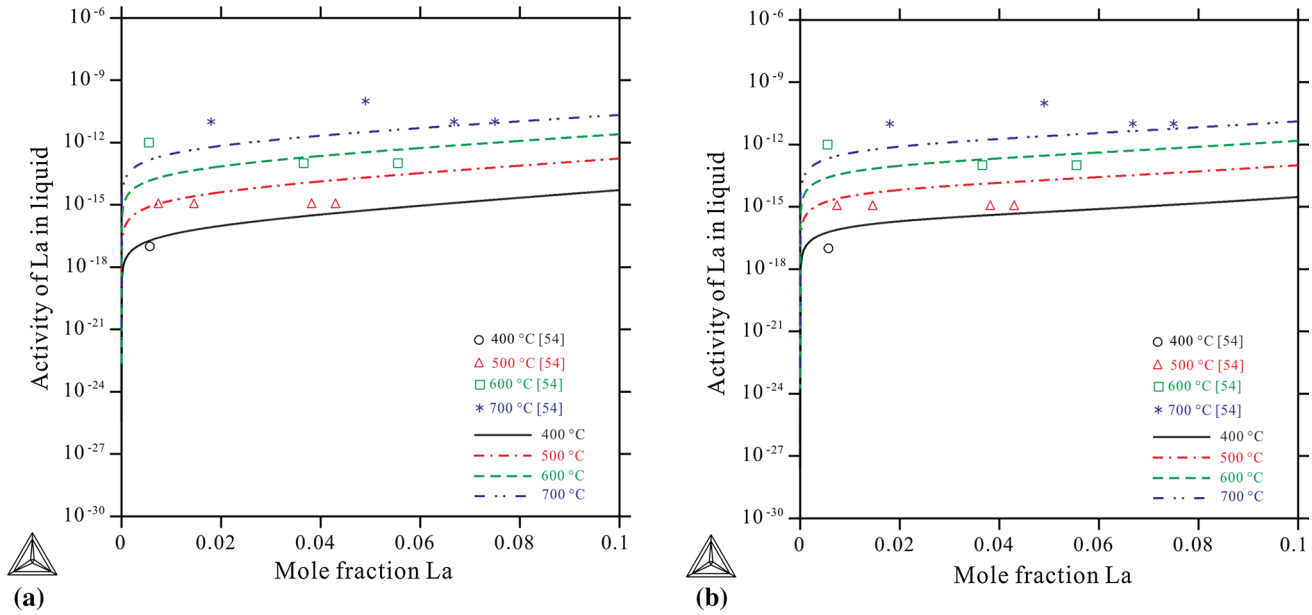

Fig. 5 Calculated activities of La in the liquid of the La-Sn system with the experimental data[54] (a) using the associated solution model (Set 1); (b) using the substitutional solution model (Set 2)

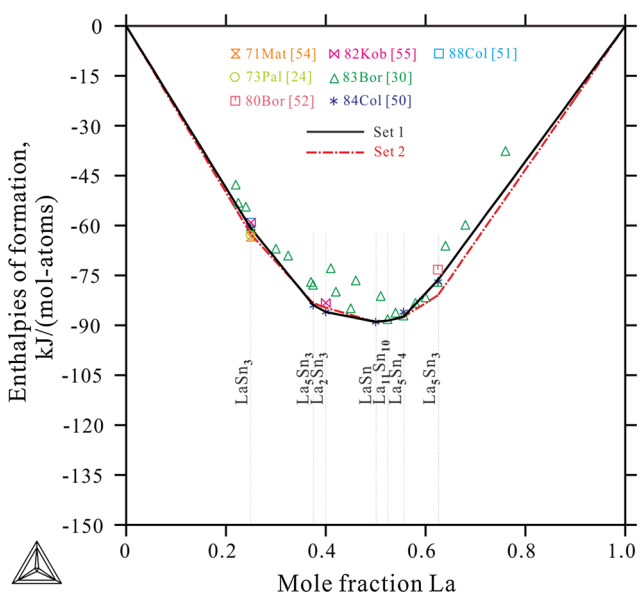

Fig. 6 Calculated enthalpies of formation of the La-Sn system for compounds in the present work using the two sets of thermodynamic parameters with the experimental data[24,30,50-52,54,55]

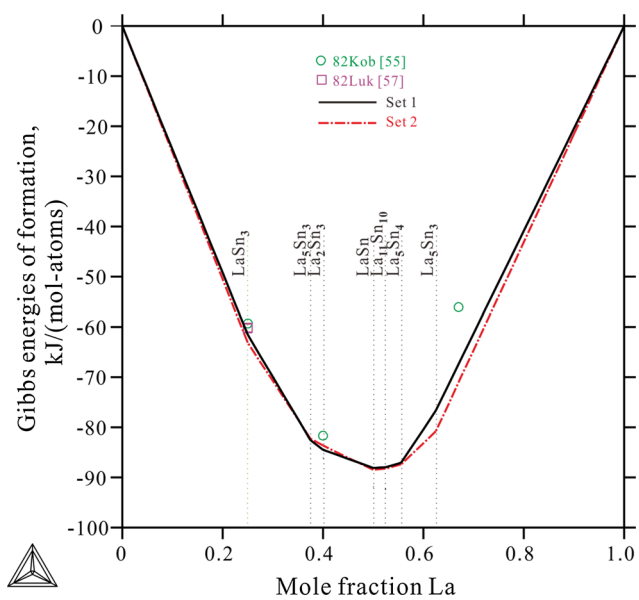

Fig. 7 Calculated Gibbs energies of formation of the La-Sn system for compounds in the present work using the two sets of thermodynamic parameters with the experimental data.[55,57] The Gibbs energies of formation data at different temperatures are converted into the same temperature, 25 °C, via the function, $G^{\mathrm{exp}}_{\mathrm{form}}(T_{\mathrm{exp}})+G^{\mathrm{cal}}_{\mathrm{form}}(25\,^{\circ}\mathrm{C})-G^{\mathrm{cal}}_{\mathrm{form}}(T_{\mathrm{exp}})$, where T_{exp} is the experimental temperature, $G^{\mathrm{exp}}_{\mathrm{form}}(T_{\mathrm{exp}})$ the measured Gibbs energies of formation value at the experimental temperature, $G^{\mathrm{cal}}_{\mathrm{form}}(25\,^{\circ}\mathrm{C})$ and $G^{\mathrm{cal}}_{\mathrm{form}}(T_{\mathrm{exp}})$ the calculated ones at 25 °C and the experimental temperature, respectively

Figure 3(a) and (b) shows the molar heat capacity and heat content of the $LaSn_3$ phase with experimental data,[44-46] respectively. From Fig. 3(a), we note that the calculated heat capacity using both sets of parameters reproduces the experimental data reasonably. The abrupt slope around 232 °C in Fig. 3(a) disappeared in the present calculation, which shows an obvious improvement compared with the previous calculations. The calculated heat content of the $LaSn_3$ phase using the presently obtained thermodynamic parameters reproduce the experimental data well, as shown in Fig. 3(b).

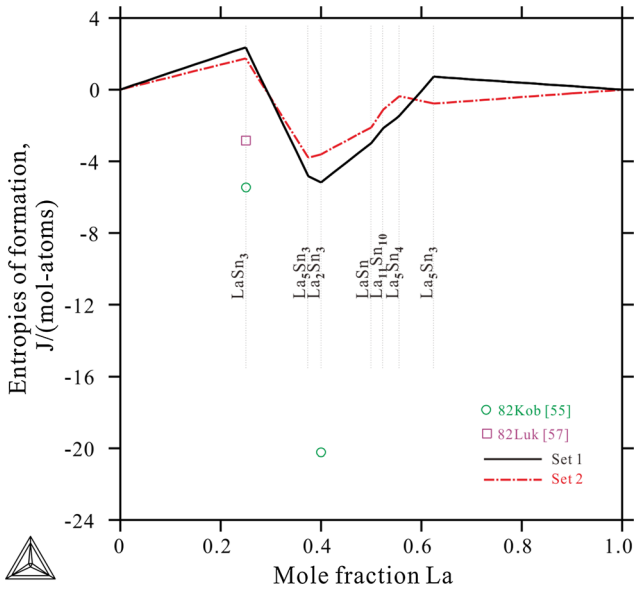

Fig. 8 Calculated entropies of formation of the La-Sn system for compounds in the present work using the two sets of thermodynamic parameters with the experimental data.[55,57] The entropies of formation data at different temperatures are converted into the same temperature, 25 °C, via the function, $S^{\mathrm{exp}}_{\mathrm{form}}(T_{\mathrm{exp}}) + S^{\mathrm{cal}}_{\mathrm{form}}(25\,^{\circ}\mathrm{C}) - S^{\mathrm{cal}}_{\mathrm{form}}(T_{\mathrm{exp}})$, where T_{exp} is the experimental temperature, $S^{\mathrm{exp}}_{\mathrm{form}}(T_{\mathrm{exp}})$ the measured Gibbs energies of formation value at the experimental temperature, $S^{\mathrm{cal}}_{\mathrm{form}}(25\,^{\circ}\mathrm{C})$ and $S^{\mathrm{cal}}_{\mathrm{form}}(T_{\mathrm{exp}})$ the calculated ones at 25 °C and the experimental temperature, respectively

Figure 4 presents the calculated partial enthalpies of mixing at 1127 °C with experimental data.[19,20,43,48-51] The calculated partial enthalpies of mixing are agreed with the calculated integral enthalpy of mixing well, as shown in Fig. 2. From Fig. 4(a) and (b), we can conclude that the present calculated results using both sets of thermodynamic parameters can reproduce the experimental data reasonably. Comparing the calculated result of the present assessment with the result by Idbenali et al., a significant improvement is observed in the present work.

Figure 5 and 6 show the calculated activities of La in liquid and enthalpies of formation of compounds with the experimental data. From the figures, we can see that the calculation result using both Set 1 and Set 2 can reproduce the experimental data reasonably.

Figure 7 presents the calculated Gibbs energies of formation at 25 °C with experimental data.[55,57] Calculated Gibbs energies for $LaSn_3$ and La_2Sn_3 phase using the thermodynamic parameters of Set 1 and Set 2 are consistent with the experimental data well. The experimental data at 67 at.% La may not reproduce well in the present work as this composition was referred to the La_2Sn phase, which may not exist in the La-Sn system.

Figure 8 shows the calculated entropies of formation at 25 °C with the experimental data.[55,57] The calculated entropies of formation for $LaSn_3$ are a little positive compared to the experimental data, but it can be accepted, as the experimental data from Kober et al.[55] and Lukashenko et al.[57] had an error of about 1.67 and 0.9 J/(mol-atoms), respectively. However, a deviation between the calculated result and the experimental data for La_2Sn_3 phase still exists although many attempts have been made.

5 Conclusion

The phase diagram and thermodynamic properties of the La-Sn system were re-assessed by the CALPHAD method. The associated solution and substitutional solution models were used to describe the liquid phase in the La-Sn system and two sets of thermodynamic parameters were thus obtained for the system. Also, calculations using both sets of thermodynamic parameters obtained in the present work can reproduce the experimental data reasonably. Compared to the previous work in the literature, the calculated thermodynamic parameters reproduce the phase diagram and thermodynamic properties more reasonably, especially for the enthalpies of mixing, partial enthalpies of mixing and heat capacity of $LaSn_3$.

Acknowledgment The financial support from the National Natural Science Foundation of China (Grant No. 51771235) is greatly acknowledged.

References

1. J.H.N. van Vucht, F.A. Kuipers, and H.C.A.M. Bruning, Reversible Room-Temperature Absorption of Large Quantities of Hydrogen by Intermetallic Compounds, *Philips Res. Refits.*, 1970, **25**, p 133-140
2. M.L. Wasz, R.B. Schwarz, S. Srinivasan, and M.P. Sridhar, Sn-Substituted $LaNi_5$ Alloys for Metal Hydride Electrodes, *Mat. Res. Soc. Proceed.*, 1995, **393**, p 237-242
3. H.H.V. Mal, K.H.I. Buschow, and A.R. Miedema, Hydrogen Absorption in $LaNi_5$ and Related Compounds Experimental Observations and Their Explanation, *J. Less-Common Met.*, 1974, **35**, p 65-76
4. K.J. Kim, G. Lloyd, A. Razani, and K.T. Feldman, Jr., Development of $LaNi_5$/Cu/Sn Metal Hydride Powder Composites, *Powder Technol.*, 1998, **99**, p 40-45
5. D. Chen, L. Chen, K. Yang, J. Guo, J. Lü, and J. Yan, Electrochemical Performance of Sn-Substituted, $LaNi_5$-Based Rapidly Solidified Alloys, *J. Alloys Compd.*, 1999, **293–295**, p 724-727
6. E.M. Borzone, M.V. Blanco, A. Baruj, and G.O. Meyer, Stability of $LaNi_{5-x}Sn_x$ Cycled in Hydrogen, *Int. J. Hydrog. Energy*, 2014, **39**(16), p 8791-8796
7. D. Chen, J. Chen, Y. Zhao, H. Huo, B. Yu, and D. Shi, First-Principles Calculations of $LaNi_{5-x}Sn_xH_y$ Intermetallics and Intermediate Phase, *Acta Metall. Sin.*, 2009, **22**(5), p 330-338

8. G.B. Olson, Genomic Materials Design: The Ferrous Frontier, *Acta Mater.*, 2013, **61**(3), p 771-781
9. Y. Liang, S. Liu, and Y. Du, Thermodynamic Assessment of the Mn-H and Mg-Mn-H Systems, *J. Phase Equilib. Differ.*, 2018, **39** (2), p 186-195
10. F. Zhang, A. Qin, S. Liu, J. Wang, and Y. Du, Phase Equilibria and Solidification Characteristics of the Al-Sc-Si Alloys, *J. Mater. Sci.*, 2016, **51**(3), p 1644-1658
11. M. Huang, X. Su, F. Yin, P. Zhang, Z. Li, and C. Chen, A Thermodynamic Assessment of the La-Sn System, *J. Alloys Compd.*, 2000, **309**, p 147-153
12. M. Idbenali, C. Servant, N. Selhaoui, and L. Bouirden, A Thermodynamic Reassessment of the La-Sn System, *Calphad*, 2009, **33**(2), p 398-404
13. J. Kim, E. Thibodeau, K. Tetley-Gerard, and I.-H. Jung, Critical Evaluation and Thermodynamic Optimization of the Sn–RE Systems: Part I. Sn-RE System (RE=La, Ce, Pr, Nd and Sm), *Calphad*, 2016, **55**, p 113-133
14. A. Palenzona and S. Cirafici, The La-Sn (lanthanum-tin) System, *J. Phase Equilib.*, 1992, **13**(1), p 42-49
15. V.N. Eremenko, M.V. Bulanova, P.S. Martsenyuk, and V.E. Listovnichii, Phase Equilibria in the Lanthanum-Tin System, *Dop. Akad. Nauk Ukr. RSR*, 1989, **2**, p 79-90
16. J.-O. Andersson, T. Helander, L. Höglund, P. Shi, and B. Sundman, THERMO-CALC & DICTRA, Computational Tools for Materials Science, *Calphad*, 2002, **26**(2), p 273-312
17. W. Cao, S.L. Chen, F. Zhang, K. Wu, Y. Yang, Y.A. Chang, R. Schmid-Fetzer, and W.A. Oates, PANDAT Software with PanEngine, PanOptimizer and PanPrecipitation for Multi-component Phase Diagram Calculation and Materials Property Simulation, *Calphad*, 2009, **33**(2), p 328-342
18. C.E. Lundin and M.J. Pool, Heats of Mixing in Praseodymium-Neodymium Solid Solutions, *J. Less-Common Met.*, 1965, **9**, p 48-53
19. J.N. Pratt and A.W.H. Morris, Heats of Solution of Some Rare-Earth Elements in Liquid Tin, *J. Less-Common Met.*, 1965, **10**, p 91-97
20. F. Sommer, J. Schott, and H.G. Krull, Temperature Dependence of Partial and Integral Enthalpies of Mixing of Liquid Rare-Earth-Tin Alloys, *J. Less-Common Met.*, 1988, **144**, p 53-63
21. G. Canneri, The Alloys of Lanthanum, *Metall. Ital.*, 1931, **26**, p 806-809
22. R. Vogel and T. Heumann, Cerium-Tin Alloys, *Z. Metall.*, 1943, **72**, p 319-328
23. R.J. Gambino, N.R. Stemple, and A.M. Toxen, Superconductivity of Lanthanum Intermetallic Compounds with the Cu_3Au Structure, *J. Phys. Chem. Solids*, 1968, **29**, p 295-302
24. A. Palenzona, Dynamic Differential Calorimetry of Intermetallic Compounds. I. Heat of Formation, and Heat and Entropies of Fusion of Rare Earth-Tin Compounds, *Thermochim. Acta*, 1973, **5**, p 473-480
25. N.G. Kulagina, A.P. Bayanov, and N.M. Kulagin, Binary tin-rare earth (La, Nd, Gd, Er, Lu) systems, *Izv. Akad. Nauk SSSR Met.*, 1985, **3**, p 211-216
26. A.T. Dinsdale, SGTE Data for Pure Elements, *Calphad*, 1991, **15**, p 317-425
27. M.L. Fornasini and F. Merlo, Compounds of the Formula M_5Sn_4 and $M_{11}Sn_{10}$ Rare Earths Intermetallic Compounds, *Atti. Accad. Naz. Lincei, Cl. Sci. Fis. Mat. Nat.*, 1971, **50**, p 186-196
28. W. Jeitschko and E. Parthe, $D8_8$ Phases of the Rare-Earth Elements with Tin and Lead, *Acta Crystallogr.*, 1967, **22**, p 551-555
29. A. Palenzona and D. Merlo, Sui composti di formula M_5X_3 formati dalle terre rare con lo Sn (on Compounds of Formula M_5X_3 Formats by Rare Earths with Sn), *Atti Accad. Naz. Lincei Rend.*, 1966, **40**, p 617-622
30. G. Borzone, A. Borsese, and R. Ferro, A Contribution to the Study of the Alloying Behavior of the Rare Earths with Tin, *Z. Anorg. Allg. Chem.*, 1983, **501**, p 199-208
31. E. Franceschi, Dimorphism of La_5Sn_3, Ce_5Sn_3 and Pr_5Sn_3 Compounds, *J. Less-Common Met.*, 1979, **66**, p 175-181
32. A. Rossi, The Crystalline Structures of $LaSn_3$ and $LaPb_3$, *Atti Accad. Naz. Lincei. Cl. Sci. Fis. Mat. Nat. Rend.*, 1933, **17**, p 839-846
33. H. Haschke, H. Nowotny, and F. Benesovsky, Perovskite Carbides with Rare Earth Metals, *Monatsch. Chem.*, 1966, **97**, p 716-717
34. I.J. McColm, N.J. Clark, and B. Mortimer, Lanthanon-Tin-Carbon Systems: The Cubic Phases Ln_3Sn and Ln_3SnC, *J. Inorg. Nucl. Chem.*, 1971, **33**, p 49-61
35. V.I. Larchev and S.V. Popova, Synthesis of a Lanthanum-tin (La_3Sn) Cubic Phase at High Pressures and Temperatures, *Fiz. Tverd. Tela (Leningrad)*, 1977, **19**, p 1463-1464
36. F. Weitzer, K. Hiebl, and P. Rogl, Crystal Chemistry and Magnetism of Neodymium Stannides Including Compounds of the Structural Series RE_nSn_{3n-2}, *J. Solid State Chem.*, 1992, **98**, p 291-300
37. I. Dürr and C. Röhr, Binary Lanthanum Stannides Close to the 1:1 Ratio-Synthesis, Crystal Structure, Chemical Bonding, *Z. Anorg. Allg. Chem.*, 2012, **638**(1), p 163-176
38. Y. Zhuang, H. Deng, J. Liu, and Q. Yao, The 673 K Isothermal Section of the La-Ni-Sn Ternary System, *J. Alloys Compd*, 2004, **363**(1–2), p 228-231
39. Y. Zhan, H. Xie, J. Jiang, Y. Xu, Y. Wang, and Y. Zhuang, The Isothermal Section of the La-Cu-Sn Ternary System at 473 K, *J. Alloys Compd.*, 2008, **461**(1–2), p 570-573
40. Y. Zhan, Y. Xu, H. Xie, Z. Yu, Y. Wang, and Y. Zhuang, The Phase Relationships in the La-Ti-Sn Ternary System at 473K, *J. Alloys Compd.*, 2008, **459**(1–2), p 174-176
41. X. Chen, Y. Zhuang, and Y. Chen, Investigation on the 773 K Isothermal Section of La-Fe-Sn Ternary Systems by X-ray Powder Diffraction, *Rare Met.*, 2010, **29**(6), p 567-571
42. J. Liang, C. Liao, Y. Du, Y. Tang, Y. Han, Y. He, and S. Liu, The Isothermal Section of the Ag-La-Sn Ternary System at 400°C, *J. Alloys Compd.*, 2010, **493**(1–2), p 122-127
43. Y.O. Esin, S.P. Kolesnikov, and V.M. Baev, Enthalpies of Formation of Liquid Binary Alloys of Aluminum and Tin with Lanthanum, *Zh. Fiz. Khim.*, 1981, **55**, p 1587-1588
44. P. Lethuillier and J. Chaussy, Magnetic-suseptibility Measurements of a Praseodymium Kondo system:$La_{1-x}Pr_xSn_3$, *Phys. Rev. B*, 1976, **13**(7), p 3132-3134
45. G.A. Costa, F. Canepa, and G.L. Olcese, Thermodynamic and Magnetic Properties of Lanthanum-Tin ($LaSn_3$), *Solid State Commun.*, 1981, **40**, p 169-172
46. S. Cirafici, F. Canepa, G.L. Olcese, and G. Costa, High-Temperature Heat Capacity of the Lanthanum-Tin ($LaSn_3$) and Cerium-Tin ($CeSn_3$) Compounds, *Solid State Commun.*, 1982, **44**, p 1507-1510
47. M.L. Fornasini, P. Manfrinetti, A. Palenzona, and S.K. Dhar, R_2Sn_3 (R=La-Nd, Sm): A Family of Intermetallic Compounds with Their Own Triclinic Structure, *Z. Naturforsch.*, 2003, **58B**, p 521-527
48. J.R. Guadagno, M.J. Pool, S.S. Shen, and P.J. Spencer, Heats of Solution and Heats of Compound Formation in the Lanthanum-Tin System, *Trans. Met. Soc. AIME*, 1968, **243**, p 2013-2014
49. R. Castanet, Enthalpies of Dissolution of Lanthanum and Cerium into Tin, *C. R. Acad. Sci. Paris*, 1984, **2**, p 5-8
50. C. Colinet, A. Pasturel, A. Percheron-Guegan, and J.C. Achard, Experimental and Calculated Enthalpies of Formation of Rare Earth-Tin Alloys, *J. Less-Common Met.*, 1984, **102**, p 167-177

51. C. Colinet, A. Bessoud, and A. Pasturel, Enthalpies of Formation of Rare Earth and Uranium Tin Compounds, *J. Less-Common Met.*, 1988, **143**, p 265-278
52. A. Borsese, G. Borzone, and R. Ferro, Thermochemistry of Binary Alloys of Rare Earths: A Comparison Between Experimental and Calculated Heats of Formation, *J. Less-Common Met.*, 1980, **70**, p 213-216
53. M.J. Pool, J.R. Guadagno, *Rep. 2410, July 1967*, Denver Research Institute, Denver, CO, 1967, p 1-19
54. N.G. Matigorova and A.P. Bayanov, Thermodynamics of $LaSn_3$ and Molten Alloys Based on tin with Lanthanum, *Zh. Fiz. Khim.*, 1971, **45**, p 2440-2442
55. V.I. Kober, I.F. Nichkov, S.P. Raspopin, Thermodynamic properties of lanthanum-tin compounds, *Izv. Vyss. Uchebn. Zaved. Tsvetn. Metall.*, 1982, **3**, p 107-108
56. A.R. Miedema, P.F. de Chltel, and F.R. de Boer, Cohesion in Alloys Fundamentals of a Semi-Empirical Model, *Physica*, 1980, **100B**, p 1-28
57. G.M. Lukashenko and R.I. Polotskaya, Thermodynamic Properties of Lanthanumtin ($LaSn_3$), *Ukr. Khim. Zh. (Russ. Ed.)*, 1982, **48**, p 357-358
58. M. Schick, A. Watson, M. Baben, and K. Hack, A Modified Neumann-Kopp Treatment of the Heat Capacity of Stoichiometric Phases for Use in Computational Thermodynamics, *J. Phase Equilib. Differ.*, 2019, **40**, p 104-114
59. B. Sundman, B. Jansson, and J.O. Andersson, The Thermo-Calc Databank System, *Calphad*, 1985, **9**, p 153-190
60. A. Iandelli, Intermetallic Compounds of the Rare Earths Metals, *Phys. Chem. Met. Alloys*, 1958, **9**, p 1-11
61. I.R. Harris and G.V. Raynor, Rare earth intermediate phases: I. Phases formed with tin and indium, *J. Less-Common Met.*, 1965, **9**, p 7-19
62. E.E. Hawnga, H. Ramsma, and M.H.V. Maaren, Oscillatory Dependence of Superconductive Critical Temperature on Number of Valency Electrons in Cu_3Au-Type Alloys, *J. Phys. Chem. Sol.*, 1970, **31**, p 2653-2662
63. A.P. Bayanev, Calculation of Formation Enthalpies of Rare Earth Compounds on the Basis of Crystal Chemical Characteristics, *Izv. Akad. Nauk SSSR Neorg. Mater.*, 1973, **9**, p 959-963
64. S. Tang, K. Zhang, and X. Xie, The Electronic Structures of $LaSn_3$ and $LaIn_3$, *J. Phys. Condens. Matter*, 1989, **1**, p 2677-2682
65. I. Dürr, M. Schwarz, and C. Röhr, The Binary Tin-Rich Lanthanum Stannides La_2Sn_5 and La_3Sn_7-A Structural and Bond Theoretical Study, *J. Solid State Chem.*, 2011, **184**, p 2220-2230
66. F.A. Spedding and A.A. Daan, *Rare-Earth Metals [Russian translation]*, Metall, Moscow, 1965
67. L. Rolla, A. Iandelli, G. Canneri, and R. Vogel, Study of Rare Earth Metals and Alloys, *Z. Metallkd.*, 1943, **35**, p 29-42

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.